

Development of RAG Architecture Based on Bloom's Taxonomy Principles for Context-Sensitive Information Retrieval

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- **Retrieval-Augmented Generation (RAG)** combines information retrieval with large language models to answer questions over long documents.
- Applications: legal analysis, medical QA, scientific literature review, financial report analysis.
- **Problem:** existing RAG systems use *fixed* retrieval parameters regardless of query complexity and lack structured cognitive annotations in their memory representations.
- This project proposes improvements to the existing SOTA framework **ComoRAG** based on cognitive frameworks:
 - 1 **Bloom's Taxonomy** - adapts retrieval depth to query complexity.
 - 2 **DSRP** - enriches memory with four cognitive dimensions.

Aim

To develop a cognitive-enhanced RAG architecture that adapts retrieval parameters based on Bloom's Taxonomy principles for context-sensitive information retrieval over long documents.

Objectives

- 1 Analyze existing RAG systems and identify limitations in handling queries of varying complexity.
- 2 Design adaptive parameter scaling based on Bloom's Taxonomy cognitive levels.
- 3 Design cognitive memory enrichment based on the DSRP framework (Distinctions, Systems, Relationships, Perspectives).
- 4 Implement both enhancements as toggleable plugins within the ComoRAG framework.
- 5 Evaluate the system on the DetectiveQA long-context reasoning benchmark.
- 6 Analyze results and identify directions for future work.

Literature Review: Comparison of RAG Methods

Method	Reference	NarrativeQA F1/EM	EN.QA F1/EM	EN.MC ACC	DetectiveQA ACC
Without RAG (GPT-4o-mini)	OpenAI	27.29/7.00	29.83/12.82	30.57	30.68
BGE-M3 + Naive RAG	Chen et al. 2024	23.16/15.10	23.71/16.24	59.82	54.54
NV-Embed-v2 + Naive RAG	Lee et al. 2025	27.18/17.80	34.34/24.57	61.13	62.50
RAPTOR + IRCoT	Trivedi et al. 2023	31.35/16.00	32.09/19.36	63.76	64.77
ComoRAG	Wang et al. 2025	31.43/18.60	34.52/25.07	72.93	68.18

- ComoRAG achieves best results across all benchmarks via its **metacognitive loop**.
- **Gap**: all systems use fixed retrieval parameters - no adaptation to query complexity.

DetectiveQA (Xu et al., 2024)

- Long-context reasoning benchmark
- 154 questions across 23 detective novels
- Average **100K tokens** per document context
- Multi-hop causal reasoning required
- Multiple-choice format (A–D)
- Evaluated on **50 samples** ($\sim 100\text{K}$ tokens each)

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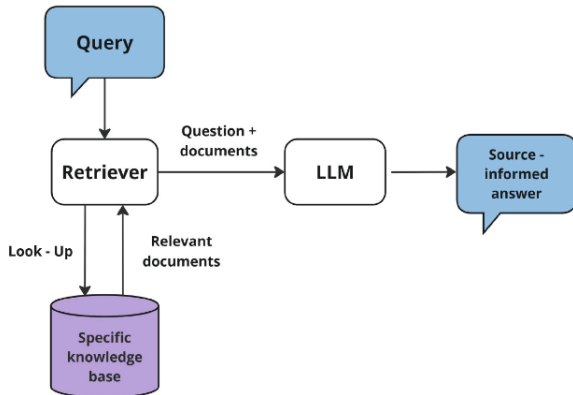
Problem Statement

Input:

- A natural language query q requiring multi-hop reasoning.
- A document context D of $\sim 100K$ tokens.

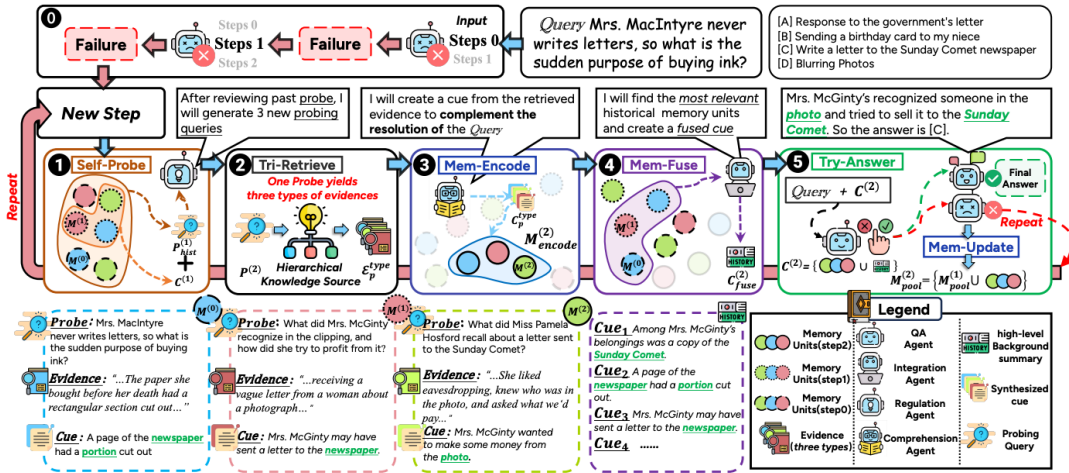
Output:

- A correct answer a to query q , grounded in evidence retrieved from D .



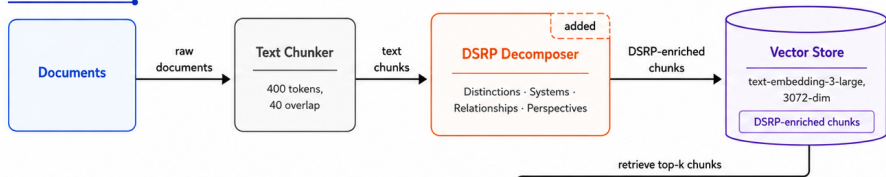
Models: ComoRAG - Base Architecture

- **Metacognitive loop:** iterative memory probing and answer refinement
- **Three-layer memory:** veridical, semantic, episodic

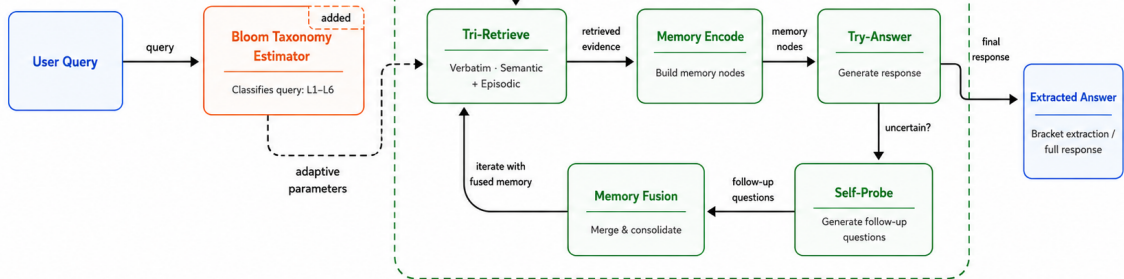


Models: Enhanced ComoRAG Architecture

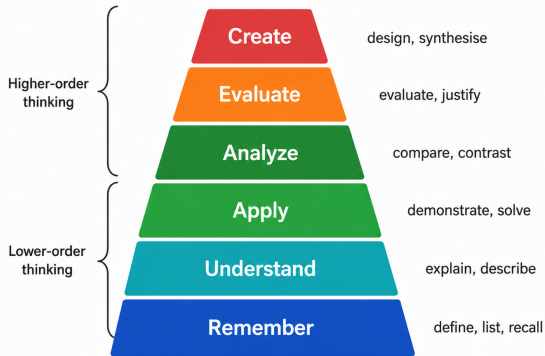
Indexing Phase



Query Phase

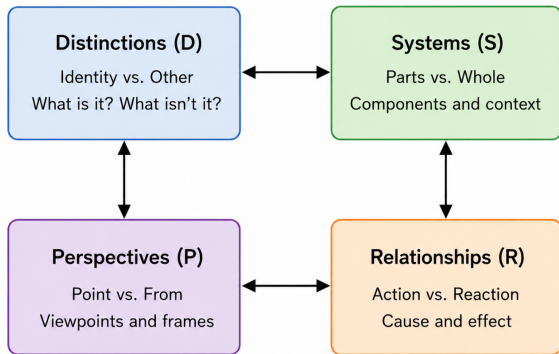


Models: Bloom's Taxonomy



Level	Name	top- <i>k</i> mult.	Iter.
1	Remember	1.0×	1
2	Understand	1.0×	1
3	Apply	2.0×	2
4	Analyze	2.0×	2
5	Evaluate	3.0×	5
6	Create	3.0×	5

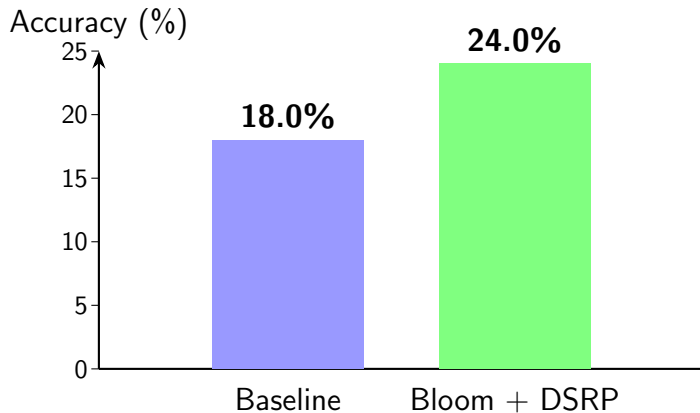
Models: DSRP



- **Distinctions** - identity vs. other
- **Systems** - part-whole relationships
- **Relationships** - cause-effect connections
- **Perspectives** - viewpoints and frames

Operates during **indexing**: annotations appended to chunk before embedding.

Results: DetectiveQA Performance



- **+6 pp** improvement on DetectiveQA (50 samples)
- **Environment:** GPT-4.1-nano, text-embedding-3-large (3072-dim)

Discussion

Why does the cognitive enhancement work?

- **Bloom Estimator**: allocates more retrieval iterations and broader top- k to higher-complexity questions (Evaluate/Create $\rightarrow 3\times$ top- k , 5 iterations).
- **DSRP Decomposer**: enriches each chunk with four cognitive annotations before embedding, improving retrieval recall for complex multi-hop questions.
- The piecewise tier design is interpretable and requires no hyperparameter tuning.

Limitations:

- **No temporal-causal modeling**: neither Bloom nor DSRP constructs cross-chunk event timelines or causal chains — questions about event order remain underserved.
- **LLM dependency**: the entire pipeline (Bloom classification, DSRP annotation, answer generation) relies on LLM reasoning quality — errors cascade across stages.
- **Limited evaluation**: evaluation was restricted to a single benchmark (50 samples) due to computational resource constraints.

Conclusion

- Analysis of existing RAG systems showed that fixed retrieval parameters limit performance on queries of varying complexity.
- A Bloom's Taxonomy plugin maps six cognitive levels to three retrieval tiers with scaled top- k and iteration counts.
- A DSRP plugin annotates chunks with Distinctions, Systems, Relationships, and Perspectives during indexing.
- Both plugins are toggleable within ComoRAG, enabling controlled A/B experimentation.
- Evaluation on DetectiveQA showed +6 pp improvement over the baseline.
- Future work: model cross-chunk temporal and causal relations, evaluate on multiple benchmarks with larger sample sizes, reduce LLM dependency with lighter classification approaches.

Thank You!

Questions?

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